PROJECT_CONTEXT.md

Project Name

Working Title: Word Chain

Elevator Pitch

Word Chain is a multiplayer word-association game.

Players build a chain by adding exactly one word to the previous player's phrase.

Example:

Player 1: water

Player 2: water bottle

Player 3: bottle cap

Player 4: cap gun

Player 5: gun powder

Each new submission must contain exactly two words.

The first word must be the second word from the previous valid submission.

The goal is to create clever, valid word connections while earning votes from other players.

Core Concept

The game is a mix of:

- word association
- creativity
- social voting
- multiplayer competition

Unlike traditional word games, players are not creating words from letters.

They are creating chains of connected concepts.

Example chain:

water water bottle bottle cap cap gun gun powder powder snow snow man man cave

The chain can continue indefinitely.

Main Rules

Rule 1

The game starts with a single word.

Example:

water

Rule 2

Every future move must contain exactly TWO words.

Example:

water bottle

Valid

Example:

water cold bottle

Invalid

Example:

bottle

Invalid

Rule 3

The first word of a player's move must match the second word of the previous move.

Example:

Previous:

water bottle

Next:

bottle cap

Valid

Next:

water cap

Invalid

Rule 4

The connection must make sense.

Examples:

bottle cap

Valid

snow man

Valid

phone charger

Valid

Examples:

bottle elephant

Likely invalid

charger banana

Likely invalid

Rule 5

No repeated phrases.

If:

water bottle

already exists in the chain,

it cannot be used again.

Rule 6

No spam.

Random keyboard input is rejected.

Example:

bottle asdfgh

Invalid

Rule 7

Players can vote.

Every submitted phrase can receive:

- Upvote
- Downvote

Community voting helps determine phrase quality.

Win Condition

Version 1:

No permanent winner.

The game continues endlessly.

Players earn points.

Leaderboard decides ranking.

Future Possibility

Timed rounds.

Highest score wins the round.

Scoring System

Suggested:

Valid Submission: +10 points

Submission Receives Upvote: +2 points

Submission Receives Downvote: -1 point

Phrase Chosen as "Best Move": +5 points

Daily Participation: +5 points

Voting System

After a phrase is submitted:

Other players can vote.

Options:

- Upvote
- Downvote

Votes affect:

- score
- ranking
- reputation

Anti-Cheat Validation

Every submission must pass:

Validation 1

Exactly two words.

Validation 2

First word matches previous second word.

Validation 3

Phrase not already used.

Validation 4

No offensive content.

Validation 5

Phrase must be meaningful.

AI can help validate.

Example:

phone charger

Pass

charger banana

Fail

Screens

Splash Screen

Shows:

- Logo
- App name

Then navigates to login.

Login Screen

Options:

- Google Login
- Guest Login

Home Screen

Buttons:

- Play
- Leaderboard
- Profile
- Settings

Shows:

- Current active chain
- Player rank

Game Lobby Screen

Shows:

- Current chain
- Number of players online
- Start word

Button:

Join Game

Game Screen

Displays:

Current Phrase:

water bottle

Input:

[bottle ____]

Player enters:

cap

System creates:

bottle cap

Submit button

Vote section

Recent moves list

Validation Screen

Success:

Move accepted

or

Move rejected

Reason shown

Leaderboard Screen

Displays:

- Global Rank
- Weekly Rank
- Top Players

Profile Screen

Displays:

- Username
- Points
- Total Moves
- Upvotes Received
- Win Statistics

Settings Screen

Displays:

- Theme
- Notifications
- Sound

UI Flow

Splash ↓ Login ↓ Home ↓ Lobby ↓ Game ↓ Results ↓ Home

Additional Routes:

Home → Leaderboard

Home → Profile

Home → Settings

Database Design

Users Table

users

id username email avatar_url points created_at

Games Table

games

id status start_word created_at

status:

active completed

Moves Table

moves

id game_id user_id first_word second_word full_phrase score created_at

Example:

first_word = bottle

second_word = cap

full_phrase = bottle cap

Votes Table

votes

id move_id user_id vote_type

vote_type:

upvote downvote

Leaderboard Cache Table

leaderboards

user_id total_points rank

AI Validation System

Purpose:

Determine whether a phrase is meaningful.

Input:

Previous: water bottle

New: bottle cap

Output:

Valid

Input:

bottle elephant

Output:

Invalid

Possible Approaches:

Phase 1

Cloud LLM API

Simple and accurate.

Phase 2

Local model

Possible future option.

Can reduce cost.

Technical Stack

Frontend

React

TypeScript

Vite

Mobile Version Later

React Native

Expo

Reason:

Single codebase.

Can target:

- Android
- iOS

Backend

Node.js

Express

Database

PostgreSQL

Authentication

Google OAuth

Guest Login

Realtime

Socket.IO

Used for:

- live chain updates
- votes
- online players

Hosting

Frontend:

Vercel

Backend:

Railway or Render

Database:

Supabase PostgreSQL

MVP Scope

Must Have:

- Login
- Create account
- Join game
- Submit phrase
- Validation
- Voting
- Leaderboard

Not Required Yet:

- Friends
- Chat
- Mobile app
- Clans
- Achievements

Future Roadmap

Phase 1

Basic Web MVP

- Login
- Single global chain
- Voting
- Leaderboard

Phase 2

Multiple Rooms

Examples:

Movies Room

Science Room

Gaming Room

General Room

Phase 3

Mobile App

React Native + Expo

Android first.

iOS later.

Phase 4

AI Improvements

Smarter validation.

Better anti-cheat detection.

Phrase quality scoring.

Phase 5

Competitive Mode

Timed matches.

Season rankings.

Rewards.

Key Design Philosophy

Simple to learn.

Difficult to master.

Encourages creativity.

Highly social.

Works on both web and mobile.

Can grow into a community-driven multiplayer word game.